

1. The objective of this assignment is to help predict the usage of bike rentals that would be available before the prediction date. It uses one day in advanced setup. The information was taken by the dataset that contained 1,095 daily observations with columns containing temperature, rain, humidity, wind speed, holiday, promotion, and rentals. Features were created around 1-2 weeks earlier. The dates were sorted out in chronological order. Also looked for duplicate records and missing values. The first 14 days of the data were removed because there was not enough previous information to create features.

There were a couple of features that were used for this assignment. The first one being weather features containing, rain, temperature, humidity and wind speed. The weather was one of the features because it can affect a lot of people who want to rent bikes. The second one being calendar, day of the week, month, day of the year, holiday and promotion. The calendar was used to better understand the different busy dates between weekdays, weekends, holidays, and seasons. The third rental history uses rental counts from 1, 7, 14, and 28 days earlier. The one-day lag represents the most recent rental activity, while the 7-day lag can capture weekly patterns. The 14-day and 28-day lags provide information about demand from similar points in previous weeks. This feature was one of the important ones because it shows us the recent rental activity and gives us info on future demand.

3. In this project, 5 different machine learning models were used. Linear regression, ridge regression, random forest, gradient boosting and support vector regression.

4. Each of the models tries to make its prediction as close to the rental numbers as possible. Linear regression tries to reduce the total prediction error. Ridge regression is almost like the first one, but instead it uses regularization to help prevent it from becoming too dependent on individual features. Random forests use many decision trees and combine their prediction. Gradient boosting model builds decision trees one after another, fixing the mistakes from pervious trees. Support Vector regression model tries to find a function that predicts the rental numbers while keeping prediction errors relatively small.

5. The models were first trained with the 2021 data since the model needs data to learn from. Then, after tested using the 2022 validation data to see how it performs with the new data.

- 2021 – Training

- 2022 – Validation

- 2023 – Final Test

6. The first model Mean Absolute Error tells us the average size of the prediction error. The second model Root Mean Square Error is almost the same as the first one, but it cares and shows more importance to bigger mistakes. It is useful because large prediction errors are important in bike rentals. R^2 measures how much of the variation in bike rentals can be explained by the model. A higher R^2 generally means that the model is explaining more of the changes in rental demand.

7.

Model	MAE	RMSE	R^2
Gradient Boosting	328.26	395.70	0.343
SVR	315.83	390.70	0.359
Ridge Regression	320.40	393.30	0.351
Linear Regression	319.89	393.61	0.350
Random Forest	334.65	407.26	0.304
Naive-Yesterday	419.39	533.47	-0.193
Seasonal Naive-7days	430.35	538.94	-0.218

The results show that all five machine-learning models performed substantially better than the two simple forecasting baselines on the reported test results. The Random Forest has the highest reported RMSE among the five machine-learning models, while the reported SVR has the lowest RMSE and highest R^2 . This shows why comparing multiple model families is useful.

8. The previous rental history can help us determine future demand. The weather features are also really important because people will probably not rent bikes during bad conditions. Lastly, the calendar features play a big role because it gives an idea of people renting during seasons, holidays, weekends and weekdays.

9. Each model has its pros and cons. The pros of the leaner regressions are easy to understand and fast to train while the cons are, could miss on complicated patterns. For Ridge regressions, the pros are similar to leaner, and the cons are mainly based on leaner regressions. Random forest the pros are works good with different features and the con is its harder to explain. Gradient boosting the pro is it's able to learn complicated patterns, and the con is that it has a bunch of settings to choose from. The pro for SVR is that it works well with complicated data, and the con is that it takes longer to train.

The actual versus predicted plot allows the predicted rental demand to be compared with the actual rental demand over time. This can help identify whether the model follows major changes and seasonal patterns in demand. The residual plot examines the difference between actual and predicted rentals. A positive residual means that the model underpredicted demand, while a negative residual means that it overpredicted demand.